LVM Session 2 Summary :

- If we want to extend the LV then we need space in that same VG.

- If we don't have space in VG and we want to extend the space of one of LV then we have 2 ways,
  1) Extend the VG using add more HD
  2) Reduce one of LV size and give back to VG and give that space to another LV.

- As we already have 2 LV of size 60 GiB and 39 GiB and we store some data in both hard disks.

```
[root@localhost ~]# df -hT
Filesystem              Type      Size  Used Avail Use% Mounted on
devtmpfs                devtmpfs  955M     0  955M   0% /dev
tmpfs                   tmpfs     984M     0  984M   0% /dev/shm
tmpfs                   tmpfs     394M  6.3M  388M   2% /run
/dev/mapper/rhel-root   xfs        47G  9.1G   38G  20% /
/dev/sda1               xfs      1014M  231M  784M  23% /boot
tmpfs                   tmpfs     197M  112K  197M   1% /run/user/0
/dev/sr0                iso9660   8.0G  8.0G     0 100% /run/media/root/RHEL-9-0-0-BaseOS-x86_64
/dev/mapper/newvg-newlv1 ext4       59G   24K   56G   1% /l1
/dev/mapper/newvg-newlv2 ext4       39G   24K   37G   1% /l2
[root@localhost ~]# cd /l1
[root@localhost l1]# cat > l1.txt
L1
[root@localhost l1]# cat l1.txt
L1
[root@localhost l1]# cd ..
[root@localhost /]# cd l2
[root@localhost l2]# cat > l2.txt
L2


[root@localhost l2]#
[root@localhost l2]#
[root@localhost l2]# cat l2.txt
L2
```

- We don't have space in VG and still we extend LV then it will show error.

```
[root@localhost ~]# vgdisplay
  --- Volume group ---
  VG Name               newvg
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  3
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                2
  Open LV               2
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               <100.00 GiB
  PE Size               4.00 MiB
  Total PE              25599
  Alloc PE / Size       25344 / 99.00 GiB
  Free  PE / Size       255 / 1020.00 MiB
  VG UUID               tI54YR-eH3M-4BMQ-jQqV-B2vs-pWnG-dX47xs
```

```
[root@localhost ~]# lvextend --size +10G /dev/mapper/newvg-newlv1
  Insufficient free space: 2560 extents needed, but only 255 available
```

- If we want to reduce space from LV then we have to follow 5 steps:
  1) If HD is online then we have to first unmount the partition using umount command.
  2) Then we have to clear/clean garbage from that partition.

```
[root@localhost ~]# umount /l1
[root@localhost ~]# e2fsck -f /dev/mapper/newvg-newlv1
e2fsck 1.46.5 (30-Dec-2021)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/mapper/newvg-newlv1: 12/3932160 files (0.0% non-contiguous), 325692/15728640 blocks
```

  3) Recreate Inode table for some space. Eg. If we have a total 60 GiB space and 50 GiB space we want and 10 GiB space we give back to VG then we need to recreate an inode table for 50 GiB.

```
[root@localhost ~]# resize2fs /dev/mapper/newvg-newlv1 50G
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/mapper/newvg-newlv1 to 13107200 (4k) blocks.
The filesystem on /dev/mapper/newvg-newlv1 is now 13107200 (4k) blocks long.
```

  4) Reduce the actual physical space from that LV and add to the VG.

```
[root@localhost ~]# lvreduce --size 50G /dev/mapper/newvg-newlv1
  WARNING: Reducing active logical volume to 50.00 GiB.
  THIS MAY DESTROY YOUR DATA (filesystem etc.)
Do you really want to reduce newvg/newlv1? [y/n]: y
  Size of logical volume newvg/newlv1 changed from 60.00 GiB (15360 extents) to 50.00 GiB (12800 extents).
  Logical volume newvg/newlv1 successfully resized.
```

```
[root@localhost ~]# lvdisplay /dev/newvg/newlv1
  --- Logical volume ---
  LV Path                /dev/newvg/newlv1
  LV Name                newlv1
  VG Name                newvg
  LV UUID                BRBY26-1iqn-49vQ-JbV0-GSho-2BRs-xRfgUx
  LV Write Access        read/write
  LV Creation host, time localhost.localdomain, 2022-11-07 12:59:17 +0530
  LV Status              available
  # open                 0
  LV Size                50.00 GiB
  Current LE             12800
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:2
```

```
[root@localhost ~]# vgdisplay newvg
  --- Volume group ---
  VG Name               newvg
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  4
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                2
  Open LV               1
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               <100.00 GiB
  PE Size               4.00 MiB
  Total PE              25599
  Alloc PE / Size       22784 / 89.00 GiB
  Free  PE / Size       2815 / <11.00 GiB
  VG UUID               tI54YR-eH3M-4BMQ-jQqV-B2vs-pWnG-dX47xs
```

5) Then again mount that partition to the folder.

```
[root@localhost ~]# mount /dev/newvg/newlv1 /l1
[root@localhost ~]# cd /l1/
[root@localhost l1]# ls
l1.txt  lost+found
[root@localhost l1]# cat l1.txt
L1
```

- After getting space into the VG now we can extend LV using the lvextend command and format that extra space using resize2fs command.

```
[root@localhost ~]# lvextend --size +10G /dev/newvg/newlv2
  Size of logical volume newvg/newlv2 changed from 39.00 GiB (9984 extents) to 49.00 GiB (12544 extents).
  Logical volume newvg/newlv2 successfully resized.
```

```
[root@localhost ~]# resize2fs /dev/mapper/newvg-newlv2
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/mapper/newvg-newlv2 is mounted on /l2; on-line resizing required
old_desc_blocks = 5, new_desc_blocks = 7
The filesystem on /dev/mapper/newvg-newlv2 is now 12845056 (4k) blocks long.

[root@localhost ~]# df -hT
Filesystem              Type      Size  Used Avail Use% Mounted on
devtmpfs                devtmpfs  955M     0  955M   0% /dev
tmpfs                   tmpfs     984M     0  984M   0% /dev/shm
tmpfs                   tmpfs     394M  6.3M  388M   2% /run
/dev/mapper/rhel-root   xfs        47G  9.1G   38G  20% /
/dev/sda1               xfs      1014M  231M  784M  23% /boot
tmpfs                   tmpfs     197M  112K  197M   1% /run/user/0
/dev/sr0                iso9660   8.0G  8.0G     0 100% /run/media/root/RHEL-9-0-0-BaseOS-x86_64
/dev/mapper/newvg-newlv2 ext4      48G   52M   46G   1% /l2
/dev/mapper/newvg-newlv1 ext4      49G   53M   47G   1% /l1
```

- In RHEL we have multiple format types such as,
  1) ext4: As we already see the resize2fs command. It will be used to both, extend and reduce space.
  2) xfs : Like resize2fs command we used xfs-growfs command in xfs format type. But here we only used this command to an extend.

- As we see in hard disks we store data in terms of sectors. Like here we store data in blocks and that block is known as Physical Extend(PE) in PV and Logical Extend(LE) in LV. By Default that PE or LE size is 4 MiB.

- If we create a LV of size 6 MiB then it will automatically create 8 MiB(Multiple of 4 MiB).

PVMove from one disk to another step by step process:

```
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda             8:0    0   20G  0 disk
sdb             8:16   0   20G  0 disk
sr0            11:0    1 1024M  0 rom
nvme0n1       259:0    0   40G  0 disk
├─nvme0n1p1   259:1    0  600M  0 part /boot/efi
├─nvme0n1p2   259:2    0    1G  0 part /boot
└─nvme0n1p3   259:3    0 38.4G  0 part
  ├─rhel-root 253:0    0 34.5G  0 lvm  /
  └─rhel-swap 253:1    0  3.9G  0 lvm  [SWAP]
[root@localhost ~]#
```

```
[root@localhost ~]# fdisk /dev/sda

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x18fc33ae.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
```

```
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943039, de

Created a new partition 1 of type 'Linux' and of size 20 GiB.

Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@localhost ~]# partprobe
```

```
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda             8:0    0   20G  0 disk
└─sda1          8:1    0   20G  0 part
sdb             8:16   0   20G  0 disk
sr0            11:0    1 1024M  0 rom
nvme0n1       259:0    0   40G  0 disk
├─nvme0n1p1   259:1    0  600M  0 part /boot/efi
├─nvme0n1p2   259:2    0    1G  0 part /boot
└─nvme0n1p3   259:3    0 38.4G  0 part
  ├─rhel-root 253:0    0 34.5G  0 lvm  /
  └─rhel-swap 253:1    0  3.9G  0 lvm  [SWAP]
[root@localhost ~]#
```

```
[root@localhost ~]# pvcreate /dev/sda1
  Physical volume "/dev/sda1" successfully created.
[root@localhost ~]# pvs
  PV             VG   Fmt  Attr PSize   PFree
  /dev/nvme0n1p3 rhel lvm2 a--   38.41g      0
  /dev/sda1           lvm2 ---  <20.00g <20.00g
[root@localhost ~]#
```

```
[root@localhost ~]# vgcreate datavg /dev/sda1
  Volume group "datavg" successfully created
[root@localhost ~]# vgs
  VG     #PV #LV #SN Attr   VSize   VFree
  datavg   1   0   0 wz--n- <20.00g <20.00g
  rhel     1   2   0 wz--n-  38.41g      0
[root@localhost ~]#
```

```
[root@localhost ~]# lvcreate -l 100%FREE -n datalv datavg
  Logical volume "datalv" created.
[root@localhost ~]# lvs
  LV     VG     Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Conver
t
  datalv datavg -wi-a----- <20.00g

  root   rhel   -wi-ao----  34.50g

  swap   rhel   -wi-ao----  <3.91g

[root@localhost ~]#
```

```
[root@localhost ~]# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0   20G  0 disk
└─sda1                8:1    0   20G  0 part
  └─datavg-datalv 253:2    0   20G  0 lvm
sdb                   8:16   0   20G  0 disk
sr0                  11:0    1 1024M  0 rom
nvme0n1             259:0    0   40G  0 disk
├─nvme0n1p1         259:1    0  600M  0 part /boot/efi
├─nvme0n1p2         259:2    0    1G  0 part /boot
└─nvme0n1p3         259:3    0 38.4G  0 part
  ├─rhel-root       253:0    0 34.5G  0 lvm  /
  └─rhel-swap       253:1    0  3.9G  0 lvm  [SWAP]
[root@localhost ~]#
```

```
[root@localhost ~]# mkfs.xfs /dev/datavg/datalv
meta-data=/dev/datavg/datalv     isize=512    agcount=4, agsize=1310464 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
data     =                       bsize=4096   blocks=5241856, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@localhost ~]#
```

```
[root@localhost ~]# mkdir /data
[root@localhost ~]# mount /dev/datavg/datalv /data/
[root@localhost ~]# df -hT
Filesystem                Type      Size  Used Avail Use% Mounted on
devtmpfs                  devtmpfs  4.0M     0  4.0M   0% /dev
tmpfs                     tmpfs     1.8G     0  1.8G   0% /dev/shm
tmpfs                     tmpfs     724M   12M  712M   2% /run
/dev/mapper/rhel-root     xfs        35G  4.3G   31G  13% /
/dev/nvme0n1p2            xfs       960M  291M  670M  31% /boot
/dev/nvme0n1p1            vfat      599M  7.0M  592M   2% /boot/efi
tmpfs                     tmpfs     362M  116K  362M   1% /run/user/1000
tmpfs                     tmpfs     362M   36K  362M   1% /run/user/0
/dev/mapper/datavg-datalv xfs        20G  175M   20G   1% /data
```

```
[root@localhost data]# touch {1..50}
[root@localhost data]# ls
1   12  15  18  20  23  26  29  31  34  37  4   42  45  48  50  8
10  13  16  19  21  24  27  3   32  35  38  40  43  46  49  6   9
11  14  17  2   22  25  28  30  33  36  39  41  44  47  5   7
[root@localhost data]# mkdir aa bb cc dd
[root@localhost data]# ls
1   12  15  18  20  23  26  29  31  34  37  4   42  45  48  50  8   bb
10  13  16  19  21  24  27  3   32  35  38  40  43  46  49  6   9   cc
11  14  17  2   22  25  28  30  33  36  39  41  44  47  5   7   aa  dd
[root@localhost data]# cal > cal.txt
[root@localhost data]# cat cal.txt
     March 2024
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
```

```
[root@localhost ~]# md5sum /data/cal.txt
71bd946348f0a6efbfa34696a8520f73  /data/cal.txt
[root@localhost ~]#
```

```
[root@localhost ~]# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0   20G  0 disk
└─sda1                8:1    0   20G  0 part
  └─datavg-datalv 253:2    0   20G  0 lvm  /data
sdb                   8:16   0   20G  0 disk
sr0                  11:0    1 1024M  0 rom
nvme0n1             259:0    0   40G  0 disk
├─nvme0n1p1         259:1    0  600M  0 part /boot/efi
├─nvme0n1p2         259:2    0    1G  0 part /boot
└─nvme0n1p3         259:3    0 38.4G  0 part
  ├─rhel-root       253:0    0 34.5G  0 lvm  /
  └─rhel-swap       253:1    0  3.9G  0 lvm  [SWAP]
[root@localhost ~]#
```

```
[root@localhost ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x5e268ad5.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943039, defaul[illegible]039):

Created a new partition 1 of type 'Linux' and of size 20 GiB.
```

```
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943039, default 41943039):

Created a new partition 1 of type 'Linux' and of size 20 GiB.

Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@localhost ~]# partprobe
```

```
[root@localhost ~]# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0   20G  0 disk
└─sda1                8:1    0   20G  0 part
  └─datavg-datalv 253:2    0   20G  0 lvm  /data
sdb                   8:16   0   20G  0 disk
└─sdb1                8:17   0   20G  0 part
sr0                  11:0    1 1024M  0 rom
nvme0n1             259:0    0   40G  0 disk
├─nvme0n1p1         259:1    0  600M  0 part /boot/efi
├─nvme0n1p2         259:2    0    1G  0 part /boot
└─nvme0n1p3         259:3    0 38.4G  0 part
  ├─rhel-root       253:0    0 34.5G  0 lvm  /
  └─rhel-swap       253:1    0  3.9G  0 lvm  [SWAP]
[root@localhost ~]#
```

```
[root@localhost ~]# pvcreate /dev/sdb1
  Physical volume "/dev/sdb1" successfully created.
[root@localhost ~]# pvs
  PV             VG     Fmt  Attr PSize   PFree
  /dev/nvme0n1p3 rhel   lvm2 a--   38.41g      0
  /dev/sda1      datavg lvm2 a--  <20.00g      0
  /dev/sdb1             lvm2 ---  <20.00g <20.00g
[root@localhost ~]# vgs
  VG     #PV #LV #SN Attr   VSize   VFree
  datavg   1   1   0 wz--n- <20.00g    0
  rhel     1   2   0 wz--n-  38.41g    0
[root@localhost ~]# pvs
  PV             VG     Fmt  Attr PSize   PFree
  /dev/nvme0n1p3 rhel   lvm2 a--   38.41g      0
  /dev/sda1      datavg lvm2 a--  <20.00g      0
  /dev/sdb1             lvm2 ---  <20.00g <20.00g
[root@localhost ~]#
```

```
[root@localhost ~]# vgextend datavg /dev/sdb1
  Volume group "datavg" successfully extended
[root@localhost ~]# vgs
  VG      #PV #LV #SN Attr   VSize  VFree
  datavg    2   1   0 wz--n- 39.99g <20.00g
  rhel      1   2   0 wz--n- 38.41g      0
[root@localhost ~]#
```

```
[root@localhost ~]# ls -l /dev/ | grep sd
brw-rw----. 1 root       disk      8,   0 Mar 24 20:08 sda
brw-rw----. 1 root       disk      8,   1 Mar 24 20:09 sda1
brw-rw----. 1 root       disk      8,  16 Mar 24 20:08 sdb
brw-rw----. 1 root       disk      8,  17 Mar 24 20:09 sdb1
[root@localhost ~]# dmsetup deps /dev/datavg/datalv
1 dependencies  : (8, 1)
[root@localhost ~]#
```

```
[root@localhost ~]# pvs
  PV             VG     Fmt  Attr PSize   PFree
  /dev/nvme0n1p3 rhel   lvm2 a--   38.41g      0
  /dev/sda1      datavg lvm2 a--  <20.00g      0
  /dev/sdb1      datavg lvm2 a--  <20.00g <20.00g
[root@localhost ~]#
```

```
[root@localhost ~]# pvmove -n datalv /dev/sda1 /dev/sdb1
  /dev/sda1: Moved: 1.21%
  /dev/sda1: Moved: 50.93%
  /dev/sda1: Moved: 100.00%
[root@localhost ~]#
```

```
[root@localhost ~]# pvs
  PV             VG     Fmt  Attr PSize   PFree
  /dev/nvme0n1p3 rhel   lvm2 a--   38.41g      0
  /dev/sda1      datavg lvm2 a--  <20.00g <20.00g
  /dev/sdb1      datavg lvm2 a--  <20.00g      0
[root@localhost ~]# vgs
  VG     #PV #LV #SN Attr   VSize  VFree
  datavg   2   1   0 wz--n- 39.99g <20.00g
  rhel     1   2   0 wz--n- 38.41g      0
[root@localhost ~]# lvs
  LV     VG     Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Conver
t
  datalv datavg -wi-ao---- <20.00g

  root   rhel   -wi-ao----  34.50g

  swap   rhel   -wi-ao----  <3.91g

[root@localhost ~]#
```

```
[root@localhost ~]# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0   20G  0 disk
└─sda1                8:1    0   20G  0 part
sdb                   8:16   0   20G  0 disk
└─sdb1                8:17   0   20G  0 part
  └─datavg-datalv 253:2    0   20G  0 lvm  /data
sr0                  11:0    1 1024M  0 rom
nvme0n1             259:0    0   40G  0 disk
├─nvme0n1p1         259:1    0  600M  0 part /boot/efi
├─nvme0n1p2         259:2    0    1G  0 part /boot
└─nvme0n1p3         259:3    0 38.4G  0 part
  ├─rhel-root       253:0    0 34.5G  0 lvm  /
  └─rhel-swap       253:1    0  3.9G  0 lvm  [SWAP]
```

```
[root@localhost ~]# vgreduce datavg /dev/sda1
  Removed "/dev/sda1" from volume group "datavg"
[root@localhost ~]# pvs
  PV             VG     Fmt  Attr PSize   PFree
  /dev/nvme0n1p3 rhel   lvm2 a--   38.41g      0
  /dev/sda1             lvm2 ---  <20.00g <20.00g
  /dev/sdb1      datavg lvm2 a--  <20.00g      0
[root@localhost ~]#
```

```
[root@localhost ~]# vgs
  VG     #PV #LV #SN Attr   VSize   VFree
  datavg   1   1   0 wz--n- <20.00g    0
  rhel     1   2   0 wz--n-  38.41g    0
[root@localhost ~]# df -hT
Filesystem                Type      Size  Used Avail Use% Mounted on
devtmpfs                  devtmpfs  4.0M     0  4.0M   0% /dev
tmpfs                     tmpfs     1.8G     0  1.8G   0% /dev/shm
tmpfs                     tmpfs     724M   12M  712M   2% /run
/dev/mapper/rhel-root     xfs        35G  4.3G   31G  13% /
/dev/nvme0n1p2            xfs       960M  291M  670M  31% /boot
/dev/nvme0n1p1            vfat      599M  7.0M  592M   2% /boot/efi
tmpfs                     tmpfs     362M  116K  362M   1% /run/user/1000
tmpfs                     tmpfs     362M   36K  362M   1% /run/user/0
/dev/mapper/datavg-datalv xfs        20G  175M   20G   1% /data
[root@localhost ~]#
```